

# Tofic David Esses

tesses@berkeley.edu | (510) 365-3722 | linkedin.com/in/toficeses | github.com/toficeses | toficeses.github.io

## Education

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### University of California, Berkeley

Aug 2025 - May 2026

*Master of Engineering in Electrical Engineering and Computer Science*

- Concentration: Physical Electronics and **Integrated Circuits**
- Courses: F25: Digital Design and Integrated Circuits, ASIC Lab, Analog Integrated Circuits, PCB Engineering  
S26: Advanced Digital Integrated Circuits, Graduate Computer Architecture, Advanced PCB Engineering

### McGill University

Sep 2020 – Dec 2024

*Bachelor of Engineering in Electrical Engineering*

Montreal, Canada

## Technical Skills

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**Programming:** Verilog, SystemVerilog, C++, Java, Python, RISC-V Assembly, Bash

**VLSI & Hardware Design:** Cadence, Genus, Innovus, Virtuoso, SVA, VCS, SPICE, Altium Designer, KiCad

## Course Projects

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### RISC-V 3-Stage CPU Design | Verilog, Cadence

- Constructed RTL hardware design of a 3-stage **RISC-V processor** and cache implementation
- Supported 39 RV32I base instruction set and applied data forwarding solving hazards, and verified with testbenches
- Developed both direct-mapped cache and 2-way set associative cache and the interface with main memory
- Floorplanned SRAMs and Core, ending up with no timing violations and DRC errors, while minimizing area

### CMOS Display Driver Design | Cadence Virtuoso

- Designed a 2-stage op-amp display driver in 45nm CMOS, achieving 70 dB gain and meeting design requirements
- Sized the transistors using the gm/Id look-up method to meet small signal and DC operating conditions
- Simulated the DC and AC response of the circuit in Cadence Virtuoso using system-level testbenches

## Hardware Industry Experience

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### Silicon Labs

May 2024 – Aug 2024

*Applications Development Internship*

Montreal, Canada

- Developed end-device applications using **C++/C, Bash** on various SoC architectures; enabling **OTA & bootloader** updates over the IEEE 802.15.4 Thread protocol, and documented the workflow for smooth customer adoption
- Implemented the first **Matter Scenes Cluster** application (CSA-IoT) and fully documented the methodology
- Built a **Bash build automation script** for streamlining **SDK version switching**, boosting team productivity

### Matrox Imaging (now part of Zebra Technologies)

May 2022 – Aug 2022

*RTL Design and Validation Internship*

Montreal, Canada

- Implement **image-processing pipelines** on **Xilinx FPGAs** using **VHDL**, contributing to the design of new frame-grabber designs in industrial inspection
- Ensured seamless communication with acquisition cameras by adhering to the **Camera Link protocol**
- Engineered **C++ automated test suites** to probe image-acquisition edge cases, leveraging **ModelSim** waveform analysis to verify digital-signal integrity and deliver **production-grade reliability**

## Research and Development

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### Berkeley Sensor & Actuator Center

Sep 2025 – May 2026

*Hardware Designer*

- Collaborated across specialties to design a full-stack ultrasound based spatial imaging device
- Spearheaded hardware development and mentored peers in printed circuit board design
- Separated **high-speed digital signals** from sensitive analog in layout of a **6-layer mixed-signal PCB**
- Designed the first iteration for ease of assembly, testing, and application programmability
- Exploring methods to **accelerate compressed sensing algorithms on FPGA** for image reconstruction

## Leadership

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### Teaching Assistant in Electronics | McGill University

Sep 2023 – Apr 2024

- Led weekly discussions, labs, and held office hours to support the learning of over 100 students in electronics